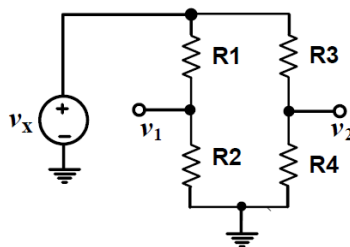
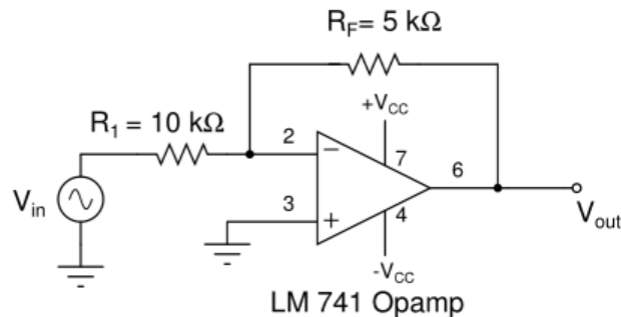


EE Dept, IIT Bombay
Academic Year: 2025-2026, Semester I (Autumn)
Course: MS101 Makerspace
EE Lecture 3&4: Operational Amplifier Circuits
Sample Problems

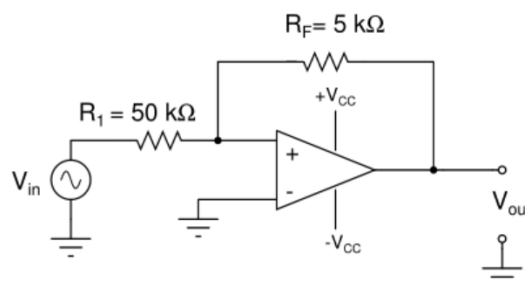
1. A Wheatstone bridge is shown below where v_x is the bridge excitation voltage and v_1 and v_2 are the output terminal voltages. Differential-mode (DM) voltage of the bridge: $v_{id} = v_1 - v_2$. Common-mode (CM) voltage of the bridge: $v_{ic} = (v_1 + v_2)/2$. Given: $v_x = 10$ V, $R_1 = R_2 = R_4 = 10$ k Ω , $R_3 = 9.5$ k Ω .
 - A) What is the bridge DM voltage in millivolts?
 - B) What is the bridge CM voltage in volts?
 - C) The output terminals v_1 and v_2 are connected to a differential amplifier with a differential gain $A_d = 20$ and a common-mode gain $A_c = 0.01$. Find the output voltage of the differential amplifier.



2. The op-amp amplifier circuit shown below has sinusoidal input voltage. Sketch on the same scale the typical waveforms of v_{in} and v_{out} .



3. A student wired the following circuit for obtaining a voltage gain (V_{out}/V_{in}) of -10 . The circuit did not work as an inverting amplifier. Rectify the mistakes in the circuit diagram to obtain a voltage gain of -10 . Do not use any components other than what is already given in the figure.



4. Choose all correct options.

An ideal op amp has finite output voltage with zero differential voltage because it has

- (A) infinite differential voltage gain.
- (B) infinite input resistances for the two inputs.
- (C) infinite common-mode rejection ratio.
- (D) dual supply voltages.

5. Mark all true statements regarding virtual short across the input terminals of an op amp.

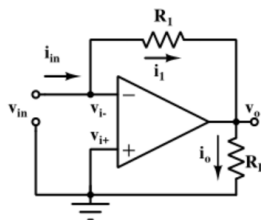
- (A) Virtual short means zero input currents into the input terminals and zero voltage across them.
- (B) Virtual short is not applicable if the op amp output reaches its higher or lower saturation voltage.
- (C) Virtual short requires the positive and negative supply voltages to have the same magnitude.
- (D) Virtual short requires the non-inverting input to be grounded.

6. Mark all correct statements.

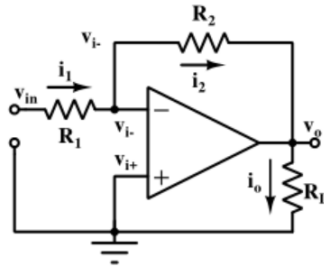
- (A) The two supply voltages in a dual-supply operational amplifier have to be equal.
- (B) The output voltage of an op amp is limited by the supply voltages.
- (C) A comparator is a differential amplifier.
- (D) Negative feedback is used in op-amp amplifier circuits to increase the output voltage range.
- (E) Negative feedback is used in op-amp amplifier circuits to get a precise gain despite large variation in the op amp's open loop gain.

7. The circuit in the figure has a load resistance R_L connected between the amplifier output and ground, and the output current i_o flows into this load. $R_1 = 15 \text{ k}\Omega$, $R_2 = 150 \text{ k}\Omega$. $R_L = 1 \text{ k}\Omega$. For this circuit,

- (A) Evaluate $R_{in} = v_{in} / i_{in}$ in $\text{k}\Omega$.
- (B) For $i_{in} = 2 \text{ mA}$, evaluate i_o in mA .



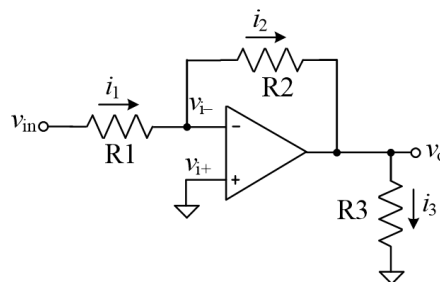
8. An op-amp circuit is shown below. $R_1 = 10 \text{ k}\Omega$, $R_2 = 50 \text{ k}\Omega$. $R_L = 2 \text{ k}\Omega$. Given: $V_{cc+} = +12 \text{ V}$, $V_{cc-} = -12 \text{ V}$. Assume that the op amp is ideal. Also assume that the maximum and minimum v_o levels are V_{cc+} and V_{cc-} respectively.



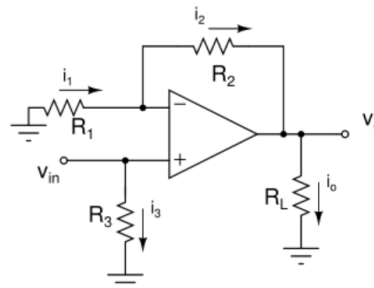
Evaluate the following:

- (A) i_o in mA for $v_{in} = +500$ mV.
- (B) Current gain A_i
- (C) Power gain A_p
- (D) Assume that $v_{in} = +500$ mV and all resistor values are as given. Now if additionally, a $10\text{ k}\Omega$ is connected between the v_{i-} and v_{i+} inputs of the op amp, what will be v_o ?

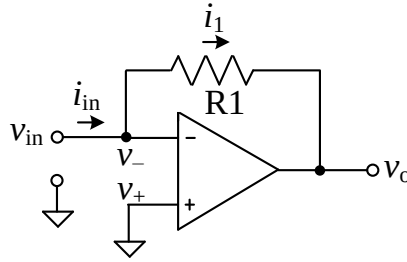
9. In the op-amp circuit shown in the figure, $R_1 = 100\text{ k}\Omega$, $R_2 = 50\text{ k}\Omega$, $R_3 = 100\text{ }\Omega$. Find the voltage, current, and power gains. Is it serving as an amplifier?



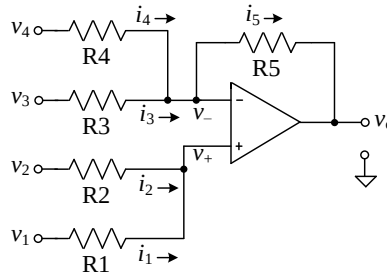
10. In the op-amp circuit shown in the figure, $R_1 = 150\text{ k}\Omega$, $R_2 = 75\text{ k}\Omega$, $R_3 = 1\text{ M}\Omega$, and $R_L = 1\text{ k}\Omega$. Find the voltage, current, and power gains.



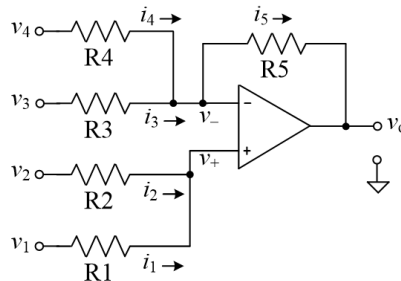
11. In the I/V converter shown in the figure, $R_1 = 100\text{ k}\Omega$. Find its trans-resistance $R = v_o/i_{in}$ and input resistance $R_{in} = v_{in}/i_{in}$.



12. In the summer circuit shown in the figure, $R_1 = R_2 = R_3 = 10 \text{ k}\Omega$, $R_4 = R_5 = 100 \text{ k}\Omega$. Find the voltage gain and input resistance for the first and third inputs, i.e. $A_{v1} = v_o/v_1$, $R_{in1} = v_1/i_1$ and $A_{v3} = v_o/v_3$, $R_{in3} = v_3/i_3$. The voltage gain and input resistance for an input is evaluated by setting all other input voltages as zero.



13. In the summer circuit shown in the figure, $R_1 = R_2 = R_3 = 20 \text{ k}\Omega$, $R_4 = R_5 = 100 \text{ k}\Omega$. Find the voltage gain and input resistance for the second and third inputs, i.e. $A_{v2} = v_o/v_2$, $R_{in2} = v_2/i_2$ and $A_{v3} = v_o/v_3$, $R_{in3} = v_3/i_3$.



14. Circuit diagram of the op-amp I/V converter used in Expt. 2 is shown below. The device under test (DUT) is connected between terminals B and C. Assume that the op amp is ideal. $+V_{CC} = +12 \text{ V}$, $-V_{CC} = -12 \text{ V}$; $R_1 = 1 \text{ k}$, $R_F = 2 \text{ k}$. Also assume that the maximum and minimum V_{Out} levels are $+V_{CC}$ and $-V_{CC}$ respectively.

